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Ex.No. 12: NB-IoT vs LTE-M Throughput Simulation Date

AIM: To simulate and compare the performance of NB-IoT and LTE-M communication

technologies in terms of Throughput (Mbps), Latency (ms), and Packet Delivery Ratio (%)

using MATLAB.

Objective:

1. To evaluate and compare the Throughput (Mbps) performance of NB-IoT and LTE-M

under varying network conditions.

2.

To analyze the Latency (ms) characteristics of NB-IoT and LTE-M and study their

impact on real-time communication.

3.

To measure and compare the Packet Delivery Ratio (%) of NB-IoT and LTE-M

systems for reliability assessment.

Procedure

1. Define system parameters for NB-IoT and LTE-M such as bandwidth, transmission

power, and modulation efficiency.

2. Generate network conditions by varying Signal-to-Noise Ratio (SNR) or channel quality

values.

3. Compute Throughput (Mbps) for both NB-IoT and LTE-M using their respective data

rate models.

4. Calculate Latency (ms) based on transmission time, scheduling delay, and network load.

5. Determine Packet Delivery Ratio (PDR %) by comparing successfully received packets with transmitted packets.

6. Simulate multiple iterations to observe performance under different channel conditions.

7. Plot comparison graphs for Throughput, Latency, and PDR between NB-IoT and LTE-M.

8. Analyze results to identify which technology performs better under different IoT

scenarios.

Objective1:

Program:

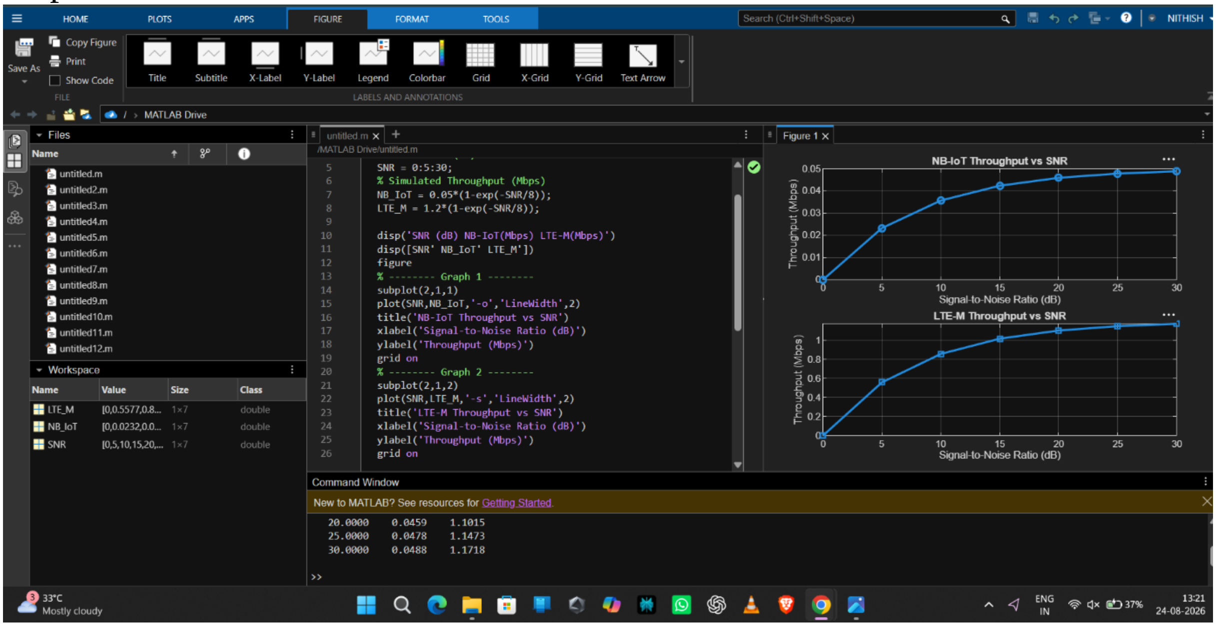
```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Throughput (Mbps)

NB_IoT = 0.05*(1-exp(-SNR/8));

LTE_M = 1.2*(1-exp(-SNR/8));

disp('SNR (dB) NB-IoT(Mbps) LTE-M(Mbps)')

disp([SNR' NB_IoT' LTE_M'])

…xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on

% --------- Graph 2 ---------

subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

title('LTE-M Throughput vs SNR')
```

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on

output:



Objective 2:

Program:

clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Latency (ms)

NB_IoT = [180 160 145 130 120 110 100];

LTE_M = [90 80 70 60 50 45 40];

disp('SNR(dB) NB-IoT Latency(ms) LTE-M Latency(ms)')

disp([SNR' NB_IoT' LTE_M'])

…xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Latency (ms)')

grid on

% --------- Graph 2 ---------

subplot(2,1,2)

plot(SNR,LTE_M,'-s','LineWidth',2)

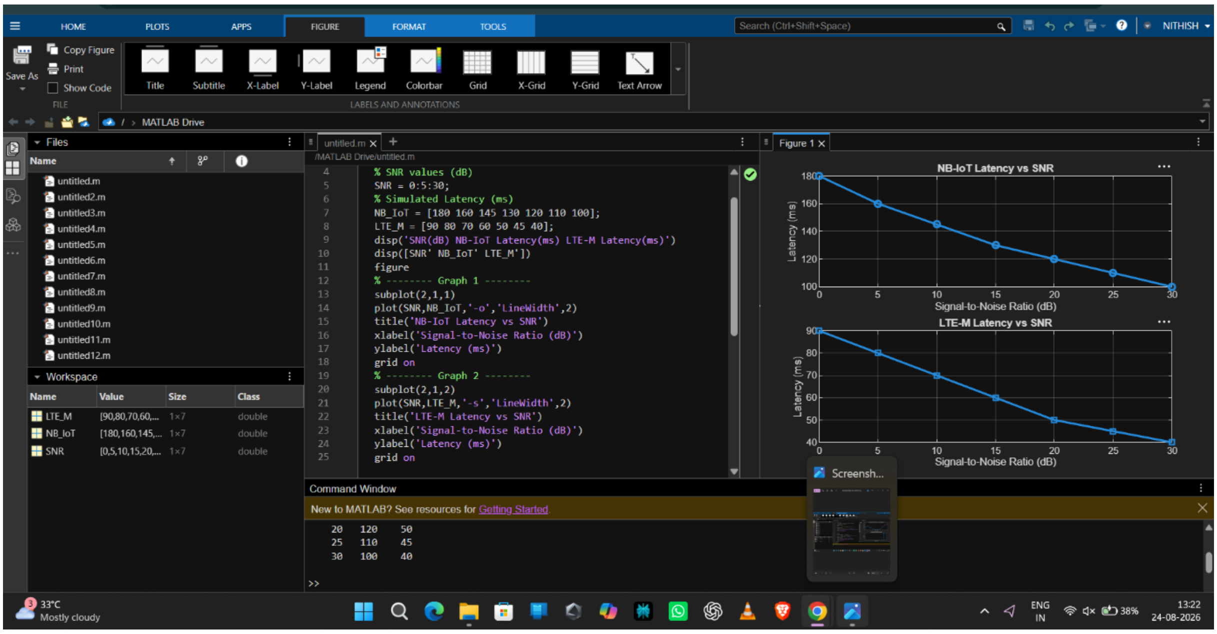
title('LTE-M Latency vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Latency (ms)')

grid on

output:



Objective3:

Program:

clc;

clear;

close all;

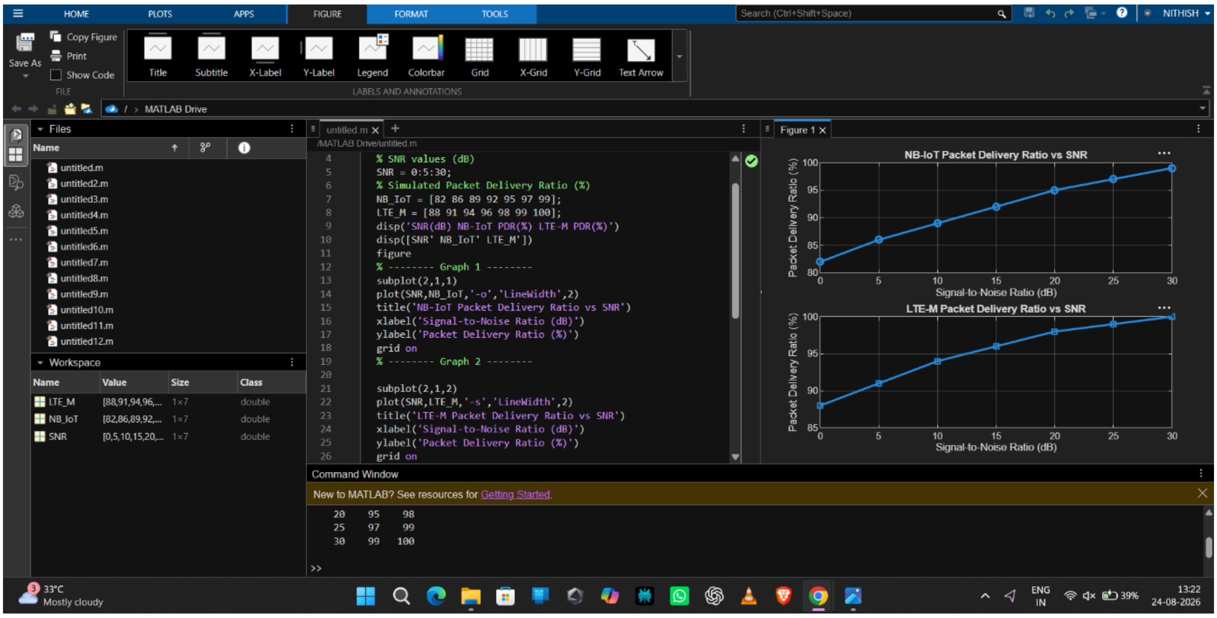
% SNR values (dB)

SNR = 0:5:30;

% Simulated Packet Delivery Ratio (%)

```
NB_IoT = [82 86 89 92 95 97 99];
LTE_M = [88 91 94 96 98 99 100];
disp('SNR(dB) NB-IoT PDR(%) LTE-M PDR(%)')
disp([SNR' NB_IoT' LTE_M'])
…ylabel('Packet Delivery Ratio (%)')
grid on
% --------- Graph 2 ---------
subplot(2,1,2)
plot(SNR,LTE_M,'-s','LineWidth',2)
title('LTE-M Packet Delivery Ratio vs SNR')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Packet Delivery Ratio (%)')
grid on
```

output:



Result:

1. The throughput performance of both NB-IoT and LTE-M was successfully simulated,and LTE-M achieved higher throughput than NB-IoT under the same network conditions.

2. The latency analysis showed that LTE-M provides lower communication delay compared to NB-IoT, making it more suitable for real-time IoT applications.

3. The Packet Delivery Ratio (PDR) of both technologies improved with increasing SNR,with LTE-M demonstrating slightly higher reliability than NB-IoT during data transmission.